

Mathematical Induction

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Where a proposition is believed to be true but a direct proof is difficult to secure, an indirect proof is sometimes obtainable by a process of reasoning called *mathematical induction*. To illustrate the process, we shall prove the following:

$$1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \cdots + n(n + 1) = \frac{n(n+1)(n+2)}{3} \quad (d)$$

Let S_n denote the sum of the terms at the left, and set $n = 1$. The left and right sides of Eq. (d) yield the following values, respectively:

$$S_1 = 1 \cdot 2 = 2 \qquad S_1 = \frac{1 \cdot 2 \cdot 3}{3} = 2$$

Thus, Eq. (d) is valid for $n = 1$. Now assume it is valid for $n = k$, where k is an arbitrary positive integer. Adding the $(k + 1)$ th term in the series and expressing it as $3(k + 1)(k + 2)/3$, we obtain

$$S_{k+1} = \frac{k(k+1)(k+2)}{3} + \frac{3(k+1)(k+2)}{3} = \frac{(k+1)(k+2)}{3}(k+3) = \frac{(k+1)(k+2)(k+3)}{3}$$

Comparing the expression for S_{k+1} with the expression in Eq. (d), we arrive at this conclusion: If Eq. (d) is valid for $n = k$, it is also valid for $n = k + 1$. We have already demonstrated that Eq. (d) is valid for $n = 1$. Therefore, it is valid for $n = 2$. Since it is valid for $n = 2$, it is also valid for $n = 3$, etc. It follows that Eq. (d) is valid for all positive integral values of n .

As a second illustration, we shall demonstrate that $x - y$ is a factor of the expression $x^n - y^n$ for all positive integral values of n . For this purpose, we write

$$x^{k+1} - y^{k+1} = x(x^k - y^k) + y^k(x - y)$$

Therefore, if the proposition is true for $n = k$, it is also true for $n = k + 1$. The proposition is true for $n = 1$, and the proof is now complete.